Gene Regulatory Network Construction in Fragile X Syndrome: An Integrative Transcriptomic, Epigenomic, and Permutation-Tested Computational Framework Across Two Independent Cohorts

Aditya Jha and Softya Sebastian

***Abstract*—Fragile X syndrome (FXS) arises from CGG-repeat expansion and promoter hypermethylation at *FMR1*, but how this silencing reorganizes the wider transcription-factor (TF) regulatory landscape, and whether small patient cohorts can detect such reorganization, is poorly characterized. We built a reproducible multi-omics pipeline combining differential expression, cell-composition-adjusted 450K methylation, and GENIE3, PANDA, LIONESS and ALPACA network inference, applied to two RNA-seq cohorts (cerebral organoids, $n = 12$; NPC/B-lymphocyte/iPSC lines, $n = 10$) and a blood methylation cohort ($n = 62$). Every network statistic was tested against a calibrated permutation null, and the power of each test was measured by simulation. *FMR1* silencing replicated in all three datasets. Beyond *FMR1*, the cohorts did not agree (effect-size $r = 0.031$), no GENIE3, PANDA or LIONESS edge survived FDR < 0.05, and ALPACA found no differential module in an exact label-permutation test (global $p = 0.83$). These tests can exclude only very large effects: at six versus six, a LIONESS edge must differ by about six standard deviations to be detected, GENIE3 detected none of 300 spiked edges even at correlation 0.99, and ALPACA missed a two-standard-deviation shift across 3% of network edges. The expression data did reshape PANDA networks beyond their priors (correlation 0.89 with shuffled-expression networks, against 0.998 between shuffles). GENIE3-predicted REST targets were enriched 4.3-fold among curated REST targets, an enrichment absent from networks built on shuffled data, whereas ChIP-seq self-recovery did not exceed random gene sets. Detecting regulatory rewiring in FXS will require substantially larger cohorts; we release the full pipeline.**

***Index Terms*—Fragile X syndrome, gene regulatory network, GENIE3, PANDA, LIONESS, ALPACA, DNA methylation, permutation testing, cross-cohort validation, tissue specificity, reproducibility.**

I. Introduction

FRAGILE X syndrome (FXS) is the most common inherited cause of intellectual disability and a leading monogenic contributor to autism spectrum disorder [1]. It results from expansion of a CGG trinucleotide repeat in the 5′ untranslated region of *FMR1* [2], which triggers promoter hypermethylation, transcriptional silencing, and loss of the fragile X messenger ribonucleoprotein (FMRP)—a process linked to broader genome instability at repeat-expansion loci [3]. FMRP normally regulates translation and localization of a large set of neuronal mRNAs, so its loss has knock-on effects across synaptic development, neuronal maturation, and cognition.

Prior transcriptomic and epigenetic work has mapped the downstream expression changes tied to *FMR1* silencing in reasonable depth [4], [5], and reversing promoter methylation partially restores neuronal phenotypes in patient-derived models [6], [7]. Less attention has gone to how the TF regulatory architecture itself is reorganized once *FMR1* is lost, and to whether that reorganization is detectable with statistical confidence in the small cohorts available for FXS.

We state our contribution precisely. The CGG-repeat $\rightarrow$ promoter-hypermethylation $\rightarrow$ *FMR1*-silencing mechanism is well established [2], [3], [6], [7], and this paper does not claim to revise it; *FMR1* silencing is reproduced here as an internal validity check on sample labels, processing and cohorts. The contribution is downstream of that mechanism and has two parts: (i) a permutation-tested, cross-cohort-validated framework for asking whether network-level regulatory reorganization after *FMR1* loss is statistically detectable at realistic FXS sample sizes, and (ii) an explicit classification of which downstream transcriptional and methylation changes are shared across tissue contexts and which are specific to one, using the three independently derived tissue sources available (cerebral organoid, NPC/B-lymphocyte/iPSC, peripheral blood) as deliberate probes rather than as a confound. We report both parts in full, including where they are negative.

Differential expression identifies genes whose mean expression changes between groups but does not model how regulatory interactions shift. Gene regulatory network (GRN) inference models TF–target relationships explicitly. GENIE3 [14] infers them from expression with tree-based ensemble regression; PANDA [15] combines expression with TF-motif and protein–protein interaction (PPI) priors by message passing; LIONESS [16] extends PANDA to sample-specific networks; and ALPACA [17] detects differential community structure

A. Jha and S. Sebastian are with Vellore Institute of Technology, Chennai, India (e-mail: aditya.jha@vitstudent.ac.in; softya.sebastian@vit.ac.in).

All code, result tables, figures, run logs and a Google Colab notebook that reproduces the analysis from public data are available at https://github.com/adj-04/fxs-research. Supplementary items S1–S15 are files in that repository (Table III).

between two networks.

We assemble these four methods into one pipeline for FXS and add three safeguards that matter at small n: (i) replication in a second, independently processed RNA-seq cohort; (ii) a permutation null for every network-level statistic, built so that its resolution is fine enough for FDR correction to be possible; and (iii) explicit adjustment for the dominant nuisance variables in each dataset (developmental timepoint in the organoids, donor in the validation lines, leukocyte composition in blood). Fig. 1 summarizes the pipeline:

1) differential expression in both RNA-seq cohorts (DESeq2), with a donor-blocked sensitivity analysis in the validation cohort;
2) cell-composition-adjusted differential methylation in blood (limma, EpiDISH);
3) GENIE3 inference of TF→target edges over a label-independent target universe;
4) PANDA networks from JASPAR motifs, STRING PPI and a continuous promoter-methylation accessibility weight;
5) LIONESS sample-specific networks with a per-edge test adjusted for timepoint;
6) ALPACA differential modules, with control as the reference network;
7) permutation nulls for GENIE3 (calibrated, pooled), PANDA (randomized prior and shuffled expression) and ALPACA (FXS/control label permutation);
8) tissue-generality classification across the three tissue contexts;
9) cross-cohort concordance (DEG overlap, effect-size correlation, rank–rank hypergeometric overlap);
10) in silico validation against ENCODE ChIP-seq, TRRUST and CollecTRI with random-set, shuffled-network and prior-only negative controls, plus donor-aware TF-activity inference in both cohorts; and
11) analytic and spike-in power estimates for LIONESS, GENIE3 and ALPACA.

We present this as a template for a rigorously validated and fully reported framework when patient data are, as is usual for a rare monogenic disorder, too few to power network inference on their own.

II. Related Work

Malachowski et al. [5] used spatial heterochromatin mapping to show coordinated silencing of long synaptic genes in FXS; an NCBI-reprocessed cohort from that study's raw data forms our validation dataset. Li et al. [4] built molecular interaction networks around FMRP-regulated transcripts with graph diffusion, focusing on RNA-binding targets rather than TF-mediated regulation. Patient-derived stem cell studies [8], [9] and mouse models [10], [11] have characterized parts of the transcriptional and neurodevelopmental phenotype, and epigenetic editing [6], [7] has established promoter methylation as central to *FMR1* silencing. Network-based disease-gene prioritization [12] and FXS mouse-brain metabolomics [13] motivate the network-centric approach.

GENIE3, PANDA, LIONESS and ALPACA are each established and are distributed together in the Network Zoo [18]. To our knowledge this is among the first frameworks to combine all four with two independent RNA-seq cohorts, FDR-capable permutation nulls for every network statistic, covariate-adjusted sample-specific networks, and three-source in silico validation, specifically for FXS.

III. Datasets

Table I lists the three public GEO datasets. Both RNA-seq cohorts use NCBI-reprocessed gene-level counts (GRCh38.p13, Entrez gene IDs) rather than submitter quantifications, so differences between them cannot be attributed to quantification pipelines. All inputs are downloaded by a script (`scripts/01_fetch_data.R`) that records file sizes and MD5 checksums (`data/MANIFEST.csv`).

For the validation cohort, NCBI reprocessed 20 of 26 submitted samples (*FMR1* full-mutation line FXS_426_NPC is absent). After excluding premutation and edited lines, 17 libraries from 10 lines and 8 donors remained; technical replicates were collapsed per line, giving 5 FXS and 5 control lines. Two donors each contribute two cell types, which we handle explicitly (Section IV-B). The methylation cohort is all male (62/62), so X-inactivation cannot confound *FMR1* promoter methylation.

The two RNA-seq cohorts differ in tissue and cell type, and the methylation cohort adds a third, non-overlapping tissue. We use this as three independent windows onto the same lesion: a gene significant in two or more contexts is more plausibly a tissue-general consequence of *FMR1* loss, and one significant in a single context more plausibly tissue-specific. This framing assumes that discordance reflects biology rather than processing differences, which we mitigate by using NCBI-reprocessed gene-level counts for both RNA-seq cohorts. The methylation cohort has no paired RNA-seq, so it indicates whether epigenetic changes are tissue-general or focal and supplies a population-level prior; it does not support subject-level multi-omic integration.

IV. Methodology

A. Implementation

The pipeline is six R scripts run in sequence, each in a fresh R session (`scripts/part1_steps1-8.R` to `scripts/part4_steps20-22.R`, `scripts/17b-run.R` and `scripts/part5_robust_nulls.R`), with shared configuration in `scripts/00_config.R`. Parts 1–4 were run on Google Colab (2 vCPU, 12.7 GB RAM) with R 4.6 and Bioconductor 3.23 installed from binary packages; Part 5 (the GENIE3, PANDA and ALPACA nulls of Section IV-I) was run from the saved Part 2 state on a local workstation with R 4.6.0, and its rerun of the observed PANDA network reproduced the Colab network exactly ($r = 1.000$). Exact package versions and session information for every stage are in `logs/`. Random seeds are fixed throughout, and every permutation loop checkpoints after each iteration.

B. RNA-seq Differential Expression

Counts were analyzed with DESeq2 [21] (Wald test; significance at adjusted $p < 0.05$ and $|\log_2 \mathrm{FC}| > 1$), using

TABLE I
DATASETS USED IN THIS STUDY.

Role	Accession	Modality	Design	Samples
Primary	GSE146339	iPSC-derived cerebral organoids (day 28, day 105)	FXS vs. Control, $\sim$ timepoint + condition	12 (6 FXS, 6 Ctrl)
Validation	GSE218676	Patient NPC / B-lymphocyte / iPSC lines	FXS vs. Control, $\sim$ cell type + condition	10 lines (5/5), 8 donors
Methylation	GSE41273	Illumina 450K array, peripheral blood (all male)	FXS vs. Control, $\sim$ condition + cell proportions	62 (9 FXS, 53 Ctrl)

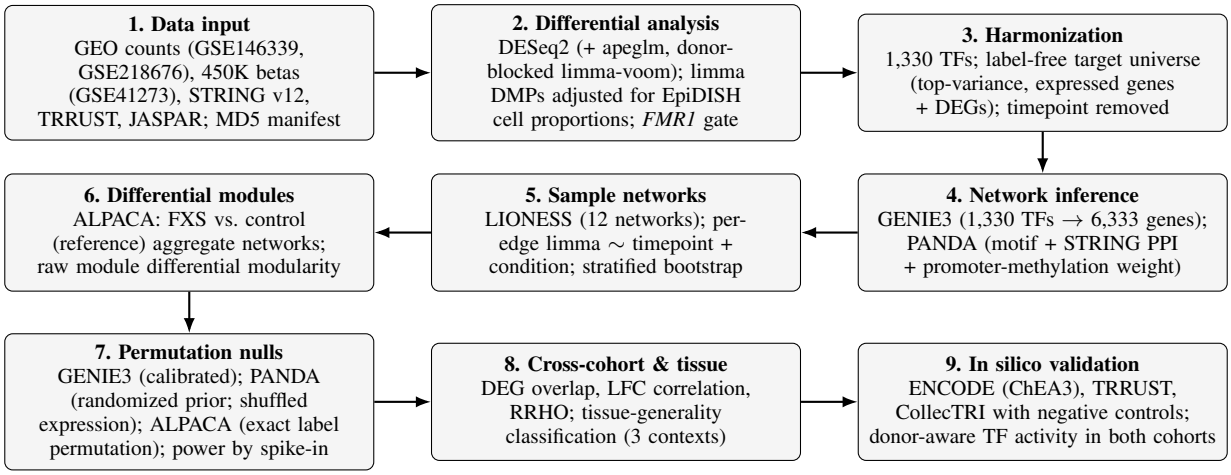


Fig. 1. Computational workflow. Each box is one stage; the flow runs left to right and snakes top to bottom. Every network-level statistic (stage 7) is tested against a permutation null before biological interpretation.

$\sim$ timepoint + condition in the primary cohort and $\sim$ cell type + condition in the validation cohort. Gene symbols were mapped from Entrez IDs with `org.Hs.eg.db`. As a check on sample labels, *FMR1* was required to be significantly downregulated in FXS before proceeding; both cohorts passed. Shrunken effect sizes (apeglm [22]) were computed alongside unshrunken ones for cross-cohort correlation. Because two validation donors contribute two lines each, we repeated the validation analysis with limma-voom [23] treating donor as a blocking factor (`duplicateCorrelation` [24]).

C. Differential DNA Methylation

Beta values for GSE41273 were taken from the GEO series matrix (485,497 probes; values verified to lie in $[0, 1]$). Leukocyte proportions (B, NK, CD4$^+$ T, CD8$^+$ T, monocytes, neutrophils, eosinophils) were estimated with EpiDISH [26] (robust partial correlation, blood reference) and included as covariates, omitting neutrophils because proportions sum to one. DMPs were called with limma [25] at adjusted $p < 0.05$ and $|\Delta\beta| > 0.1$; the unadjusted model ($\sim$ condition) is reported as a sensitivity analysis. Probes were annotated with the IlluminaHumanMethylation450k annotation via minfi [27], expanded to all annotated genes, and summarized as gene-level mean $\Delta\beta$.

D. TF List, Timepoint Adjustment and Target Universe

A curated list of 1,390 human TFs (`data/tf_list.txt`) was intersected with the expression matrix, retaining 1,330 TFs. In the primary cohort, developmental timepoint dominates the first principal component ($|r| = 0.96$ with PC1); because co-expression driven by timepoint would otherwise be read as regulation, timepoint was regressed out of the variance-stabilized expression matrix used for all network inference (limma `removeBatchEffect`, protecting the condition effect), reducing $|r|$ to 0. Differential expression was unaffected, since DESeq2 models timepoint directly.

Candidate targets were chosen without using FXS/control labels: non-TF genes with at least 10 normalized counts in at least 3 samples (18,107 of 22,011), ranked by variance, of which the top 5,000 were kept, together with all DEGs. The GENIE3 universe therefore comprised 6,333 genes (1,330 TFs, 5,003 non-TF targets; 22.1% of the 28,679-gene matrix). PANDA's motif target pool was drawn from the same ranked list (3,000 non-TF genes), so neither network's gene selection depends on the contrast later tested. Six TFs previously reported in the FXS literature—SP1, NRF1, REST, FOXG1, SIX3 and TFAP2A—were fixed a priori as reference regulators and used only to examine the recovered networks.

E. GENIE3

GENIE3 [14] was run with the 1,330 TFs as candidate regulators of all 6,333 genes (1,000 trees per target). Its weight matrix is indexed regulator $\times$ target and normalized per target; we verified both properties in the output (per-target weight sums equal to 1). Edges with zero weight and self-edges were

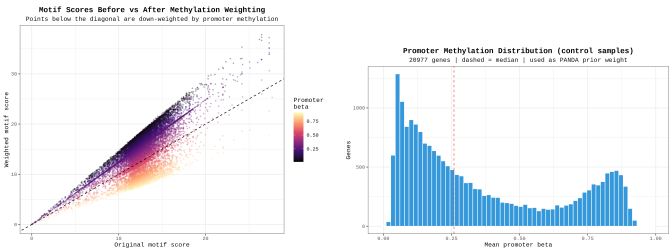


Fig. 2. Methylation-derived accessibility prior. Left: PANDA motif scores before vs. after reweighting by $1.5 - \beta_{\text{promoter}}$; points below the diagonal are down-weighted by promoter methylation. Right: distribution of control-group promoter methylation used to derive the weight.

discarded, leaving 8,260,380 TF→gene edges. Hub TFs were ranked by weighted out-degree.

F. PANDA

A motif prior was built by scanning hg38 promoters ($-2{,}000$ to $+500$ bp) of the 2,802 target genes with annotated promoters against JASPAR 2020 [20] motifs (523 TFs with motifs; 373,665 motif edges). The PPI prior used high-confidence STRING v12.0 [19] interactions (combined score $\geq$ 700; 473,860 edges); STRING protein identifiers were mapped to gene symbols with STRING's own protein-information table, which mapped all 16,201 proteins and yielded 6,490 TF–TF interactions. Motif-prior edges were reweighted by a continuous function of control-group promoter methylation, $1.5 - \beta_{\text{promoter}}$ (74.9% of targets had promoter probes; the remainder received the median), so that the prior encodes baseline promoter accessibility rather than the FXS contrast; this modulated 93.7% of motif edges (Fig. 2). PANDA [15] (union mode) converged on 873 TFs and 3,325 genes.

G. Sample-Specific Networks: LIONESS

LIONESS [16] in PANDA mode produced one network per primary-cohort sample (12 networks, 2,902,725 edges each). Differential edges were tested by fitting each edge's 12 sample-specific weights with limma, $\sim$ timepoint + condition, with Benjamini–Hochberg (BH) correction. As a genome-wide summary, we computed the proportion of edges stronger in the FXS mean than in the control mean, with a 95% bootstrap interval (1,000 resamples, stratified by timepoint within each group).

H. Differential Modularity: ALPACA

FXS and control aggregate networks were the means of the corresponding LIONESS networks. ALPACA [17] detects communities in a reference network and then finds the partition of TFs and genes that maximizes the differential modularity of the second network relative to it; the control network was the reference. ALPACA's reported node scores are normalized to sum to 2 within each module, so they rank nodes within a module but say nothing about how strongly the module differs; a module's mean node score is simply $2/\text{size}$. We therefore scored each module m by its raw differential modularity Q_m, the sum of the differential modularity matrix over the module's TF–gene pairs, together with the total $Q = \sum_m Q_m$ and a size-adjusted density $Q_m/(n_{\text{TF}}\, n_{\text{gene}})$ for modules with at least five nodes. The largest modules were exported to Cytoscape [28] through RCy3 [29] for visualization (`scripts/14_cytoscape_local.R`). The local centrality of each reference TF was its weighted out-strength over positive PANDA edges within its module, expressed as a percentile among the module's TFs.

I. Pooled Permutation Nulls

Each network statistic was compared with a null distribution generated by repeating the inference step under a null model.

GENIE3. Each gene's expression vector was shuffled independently across samples, destroying co-expression while preserving marginals (GENIE3 does not use labels, so label permutation would be uninformative). Five permutations were run with the same 1,000 trees per target as the observed network, since fewer trees give noisier, wider null weights and a miscalibrated test.

PANDA. Two nulls were used. (a) Randomized prior: the motif prior was rewired with degree preservation and its scores shuffled (20 runs), and all observed edges were compared with all null edges. (b) Shuffled expression: PANDA was rerun with the real motif and PPI priors but with each gene's expression permuted across samples (5 runs). Comparing the observed network with these, and the shuffled-expression networks with each other, measures how much the measured co-expression changes the network beyond what the priors alone produce.

ALPACA. FXS/control labels were permuted within timepoint, keeping three of six samples per timepoint labeled FXS; this gives $\binom{6}{3}^2 = 400$ labelings, 399 besides the observed one. For each relabeling, the group networks were recomputed from the LIONESS networks and the full ALPACA partition rerun. All 399 relabelings were run, so the test is exact (smallest attainable $p = 1/400 = 0.0025$). Global significance compared the observed total Q with its relabeling distribution. Module significance used the maximum statistic: each observed Q_m (and density) was compared with the largest module value in each relabeled run, which controls the family-wise error rate and accounts for selecting the strongest module.

A per-edge null—comparing each edge only with its own permutation values—cannot support FDR control at this scale: with B permutations the smallest attainable p is $1/(B+1)$ (0.048 for $B = 20$), whereas BH correction over $\sim 10^6$–10^7 edges requires p-values several orders of magnitude smaller, so "zero edges at FDR < 0.05" would be guaranteed regardless of the data. We therefore compared each observed score x with the pooled null of all edges (or nodes) across all permutations, $p = (\#\{\text{null} \geq x\} + 1)/(N + 1)$, accumulated in a constant-memory histogram (200,000 bins; a bin straddling x is counted as exceeding it, so p is conservative). This assumes scores are comparable across edges, which holds for GENIE3 (weights normalized per target over a common regulator set) and PANDA (z-scaled scores). Smallest attainable p was 2.4×10^{-8} (GENIE3) and 1.7×10^{-8} (PANDA). For a single edge, BH at

FDR 0.05 requires $p \leq 0.05/m$, which is 6×10^{-9} for the 8.26 million GENIE3 edges; the GENIE3 test can therefore reject only if at least four edges reach the null floor, and the PANDA test ($0.05/m = 1.7 \times 10^{-8}$) only at the floor. Calibration was checked by leave-one-out: each null replicate in turn was scored against the pooled null of the remaining replicates, and a calibrated test assigns $p < 0.05$ to about 5% of these held-out null values. Edge-level p-values were BH-corrected.

J. Power Analysis

For LIONESS, the minimum detectable effect (MDE) of each edge was computed from its moderated residual variance in the edge-level limma model: the condition difference giving 80% power at the BH threshold reached when k of the m edges carry true effects ($p^* = 0.05k/m$), with $k = 1$, 100 and 10,000, expressed in units of the edge's across-sample standard deviation; the same calculation was repeated for hypothetical balanced designs of 12 to 100 samples per group with unchanged per-edge variance. For GENIE3, 50 randomly chosen non-TF targets were replaced by vectors with a set correlation r (0.5 to 0.99) to a randomly chosen TF, keeping each target's mean and variance; GENIE3 was rerun for these targets with 1,000 trees, and each spiked edge was tested against the pooled null with BH correction over all edges. For ALPACA, the FXS LIONESS networks of a random block of TFs and genes were raised by δ standard deviations of the control network's edge scores, and the label-permutation test was rerun with 20 relabelings per setting; blocks of 900, 18,000 and 90,000 edges were used.

K. Tissue-Generality Classification

Each gene significant in at least one tissue context (primary DEG, validation DEG, or DMP-annotated gene) was called tissue-general if significant in at least two of the three contexts and tissue-specific if significant in exactly one. Methylation gene names from the hg19-era 450K annotation were mapped to current symbols (e.g. *ASFMR1* → *FMR1-AS1*), and probes in the shared *FMR1*/*FMR1-AS1* promoter count towards both. Genes significant in no context were excluded. For each pair of contexts, the overlap was compared with its hypergeometric expectation on the shared gene background. Because contexts differ in sample size and modality, the classification is descriptive.

L. Cross-Cohort Validation

The primary and validation cohorts were compared on the 26,014 genes testable in both: (i) DEG overlap (hypergeometric and Fisher's exact tests); (ii) genome-wide effect-size correlation (Pearson, Spearman) and sign concordance, on raw and apeglm-shrunken $\log_2$FC; and (iii) rank–rank hypergeometric overlap (RRHO) [30] of genes ranked by signed $-\log_{10} p$, on a 99×99 grid of rank cutoffs using our implementation of the original algorithm.

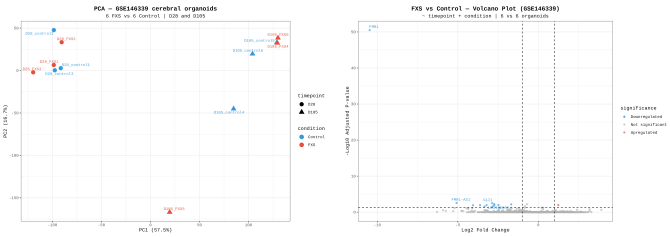


Fig. 3. Differential expression, GSE146339. Left: PCA (PC1 separates day 28 from day 105). Right: volcano plot, FXS vs. Control.

M. In Silico Regulator Validation

For candidate TFs (top hubs plus the reference TFs with database coverage), the top 200 (ENCODE) or 500 (TRRUST, CollecTRI) predicted targets from each network were tested against three independent sources: self-recovery rank among ENCODE ChIP-seq regulator libraries [32] returned by ChEA3 [31]; and hypergeometric enrichment among curated targets in TRRUST v2 [33] and CollecTRI [34] (43,159 interactions, 1,186 TFs; retrieved 9 November 2026), with BH correction. Three negative controls were applied. ENCODE ranks were compared with those from 30 random 200-gene sets drawn from the same gene universe (GENIE3 or PANDA targets), since one query ranks every ENCODE TF. GENIE3 results were repeated with the TF's top targets in each of the five gene-shuffled GENIE3 networks of the null. PANDA results were repeated with the TF's top targets in the motif prior alone, which measures how much of the recovery the prior supplies. Separately, TF activity was inferred in each cohort from gene-level statistics and CollecTRI regulons with a univariate linear model (decoupleR [35]; minimum regulon size 5), giving a TF-level cross-cohort comparison that does not require individual target genes to replicate. Organoid activity used the DESeq2 Wald statistics; validation activity used both the DESeq2 Wald statistics and the donor-blocked limma-voom moderated t-statistics, since two donors contribute two lines each (CollecTRI re-retrieved 10 November 2026 for this and the negative controls).

V. Results

A. Differential Expression and Methylation

DESeq2 identified 13 DEGs in the primary organoid cohort (12 down, 1 up), led by *FMR1* ($\log_2$FC $= -10.47$, $p_{\text{adj}} = 3.1\times10^{-51}$) and *FMR1-AS1* ($-5.08$, $p_{\text{adj}} = 2.7\times10^{-3}$); the remainder include Hedgehog and patterning genes (*SHH*, *GLI1*, *PITX2*, *HOXA2*, *HOTAIRM1*; Figs. 3, 4). In the validation cohort, 63 genes were significant with lines treated as independent, and *FMR1* replicated ($\log_2$FC $= -8.91$, $p_{\text{adj}} = 7.2 \times 10^{-8}$). The donor-blocked analysis, however, estimated a within-donor correlation of 0.66 and retained only one significant gene, *FMR1* ($\log_2$FC $= -9.19$, FDR $= 0.031$), although effect sizes agreed closely between the two models (Pearson $r = 0.84$). The 63 validation DEGs should therefore be read as nominal at the line level; *FMR1* is the only validation DEG robust to donor structure.

In blood, estimated CD8$^+$ T-cell and eosinophil proportions differed between FXS and controls (Wilcoxon $p = 0.013$ and

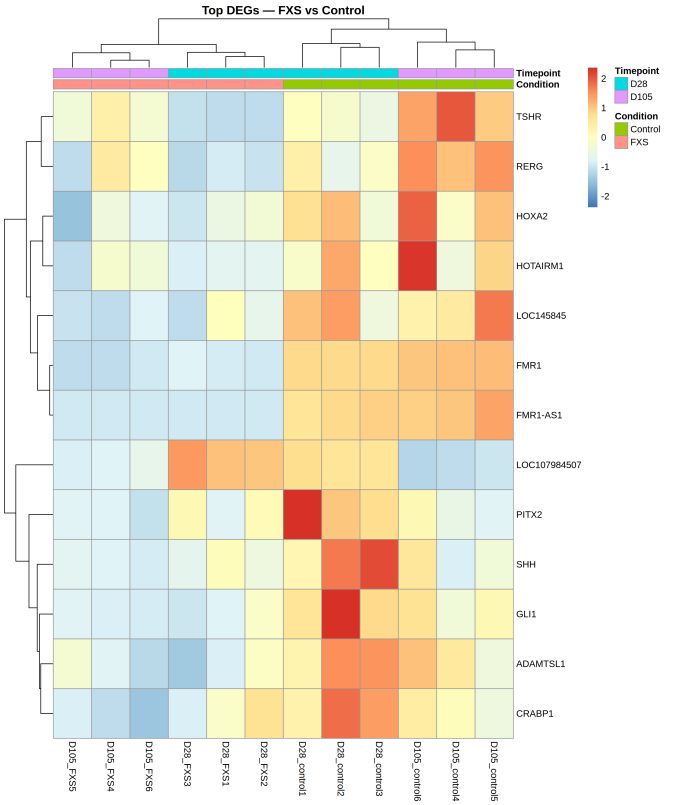


Fig. 4. The 13 differentially expressed genes, GSE146339, led by *FMR1* and *FMR1-AS1*.

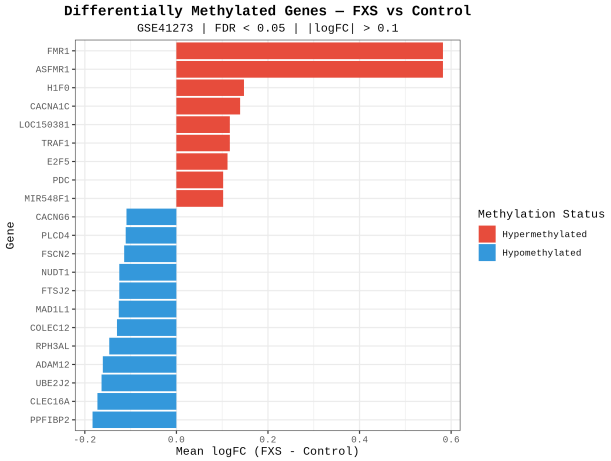


Fig. 5. Differentially methylated genes, GSE41273 (cell-composition-adjusted; FDR < 0.05, $|\Delta\beta| > 0.1$). *FMR1* and *ASFMR1* (*FMR1-AS1*) share 15 promoter probes.

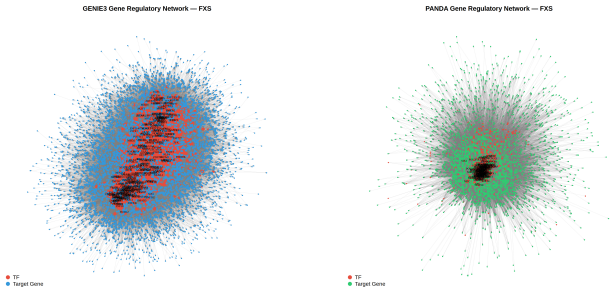


Fig. 6. Inferred TF regulatory networks. Left: GENIE3 (expression only; top 1% of edges). Right: PANDA (motif + STRING PPI + methylation prior; top decile).

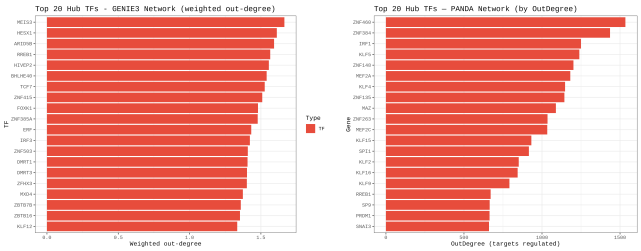


Fig. 7. Top 20 hub TFs. Left: GENIE3, weighted out-degree. Right: PANDA, out-degree. Among the reference TFs, REST is in the GENIE3 top 50 and TFAP2A in the PANDA top 50.

0.002), confirming that composition adjustment was needed. The adjusted model identified 41 DMPs (FDR < 0.05, $|\Delta\beta| > 0.1$) in 21 genes (9 hyper-, 12 hypomethylated); the unadjusted model reproduced the 25 DMPs of a composition-naive analysis. *FMR1*/*FMR1-AS1* dominated under both models (mean $\Delta\beta = 0.582$ adjusted, 0.562 unadjusted, 15 shared promoter probes), while every other DMP gene had $|\Delta\beta|$ between 0.10 and 0.18 (Fig. 5). Hypermethylation is thus strongly focal at the causal locus, as expected; we treat this as an internal validity check rather than a new mechanistic finding.

B. GENIE3 and PANDA Networks

For visualization and hub ranking, we retained the top 1% of GENIE3 edges (82,604 edges; 6,168 nodes, 1,330 TFs) and the top decile of positive PANDA edges (100,968 edges; 3,245 nodes, 518 TFs; Fig. 6). Top GENIE3 hubs by weighted out-degree were MEIS3, HESX1 and ARID5B; top PANDA hubs were ZNF460, ZNF384, IRF1 and KLF5 (Fig. 7). Of the six reference TFs, REST ranked 29th of 1,330 TFs in GENIE3 and TFAP2A 48th of 518 in PANDA; the others ranked outside the top 50.

C. Permutation Nulls

The GENIE3 null was calibrated: held-out null edges received $p < 0.05$ in 4.97% and $p < 0.01$ in 1.00% of cases (mean over five held-out replicates; 41.3 million null values pooled). Observed edges did not depart from it: 5.13% of 8,260,380 edges had $p < 0.05$, and none survived FDR < 0.10 (Fig. 8, left).

Against the randomized-prior null, with all edges on both sides, 5.17% of the 2,902,725 observed PANDA edges exceeded the null 95th percentile (3.55), the fraction expected by chance, and none survived FDR < 0.05. The two distributions were not identical (maximum CDF difference 0.19): observed scores had a heavier lower tail (22.7% below the null 5th percentile) and fewer extreme high scores (99th percentile 4.40 vs. 5.19; Fig. 8, right). The shuffled-expression null separates the contributions of priors and data (Fig. 9). With

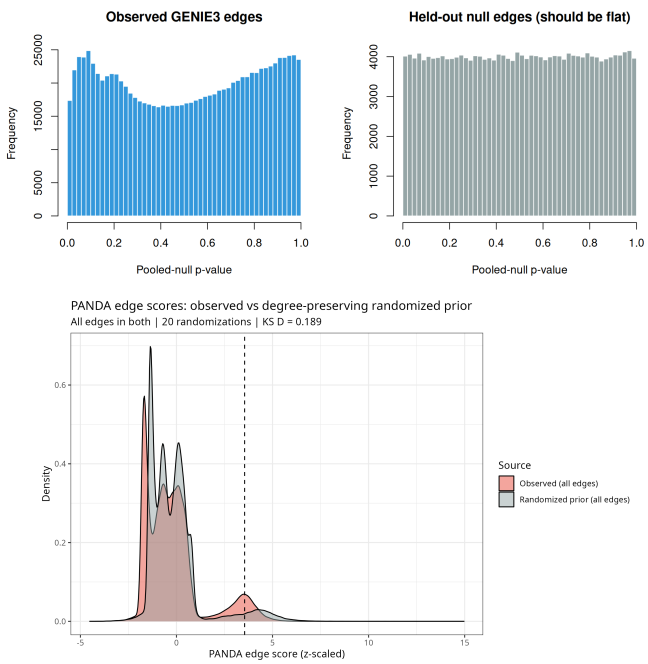


Fig. 8. Permutation nulls for edges. Top: GENIE3 pooled-null p-values for observed edges (left) and for held-out null edges (right); both are flat, so the test is calibrated and the observed edges carry no excess of small p-values. Bottom: all observed PANDA edge scores vs. all scores from the degree-preserving randomized-prior null; the upper tails match (5.17% of observed edges above the null 95th percentile, dashed), while the observed distribution has a heavier lower tail.

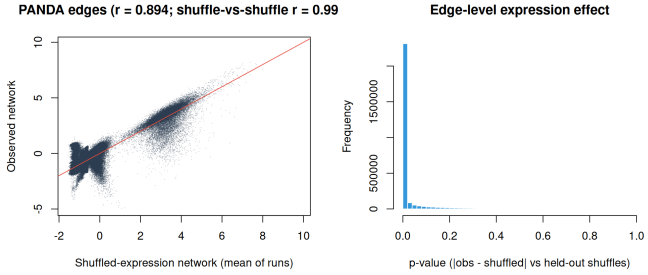


Fig. 9. PANDA with shuffled expression. Left: observed edge scores against the mean of five networks built from the same priors with each gene's expression permuted (red: identity). Shuffled-expression networks agree with each other almost perfectly ($r = 0.998$) but with the observed network less closely ($r = 0.894$), so the real expression data reshape the network. Right: per-edge p-values for the difference; because the shuffled networks barely vary, this test reflects PANDA's sensitivity to co-expression structure rather than sampling uncertainty and is not used to call edges.

shuffled expression, PANDA returned almost the same network every time (mean correlation between runs 0.998), so with uninformative expression its output is fixed by the priors. Real expression moved the network away from both: its correlation with the shuffled-expression networks was 0.894, its correlation with the motif prior fell from 0.943 to 0.846, and edge scores changed by 0.51 z-units on average. The expression data therefore contribute measurably to PANDA network structure, but not in a way that produces edges stronger than prior-derived scores.

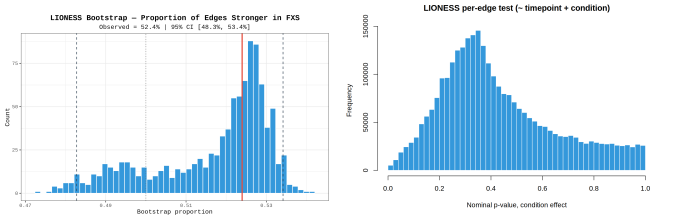


Fig. 10. LIONESS. Left: stratified bootstrap distribution of the proportion of edges stronger in FXS (95% CI includes 50%). Right: per-edge limma p-values for the condition effect ($\sim$ timepoint + condition).

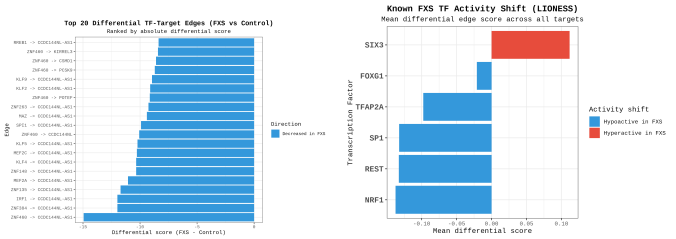


Fig. 11. LIONESS detail. Left: edges with the largest FXS-vs-control weight differences. Right: mean differential edge score of the six reference TFs across all targets.

D. Sample-Specific Networks

In the per-edge LIONESS test adjusted for timepoint, 25,147 of 2,902,725 edges (0.87%) had nominal $p < 0.05$ and none survived FDR < 0.05. The p-value distribution was not uniform but peaked near 0.3 with a deficit near zero (Fig. 10), indicating a conservative test and no excess of small p-values. Genome-wide, 52.4% of edges were stronger in FXS than control (stratified bootstrap 95% CI 48.3–53.4%), an interval that includes 50%: no systematic shift in regulatory strength was detected. Among reference TFs, mean LIONESS edge scores were lower in FXS for NRF1, REST, SP1 and TFAP2A and higher for SIX3 (Fig. 11), descriptive differences that individual edges do not support at FDR.

E. ALPACA Differential Modules

With the control network as reference, ALPACA partitioned 3,995 nodes into 26 modules, 17 with at least five nodes. Most of the differential modularity lay in one module (module 16: 375 TFs and 607 genes; $Q_m = 0.116$ of a total $Q = 0.170$), followed by module 2 (183 TFs, 658 genes; $Q_m = 0.020$). Neither the total nor any module exceeded what arbitrary groupings of the same samples produce: across all 399 within-timepoint relabelings the median total Q was 0.176 (observed 0.170, $p = 0.83$), the best module's family-wise p was 0.37 (module 16), the best size-adjusted density's family-wise p was 0.35, and no module reached $p < 0.05$ (Fig. 12). At $n = 6$ per group, FXS/control differences in community structure are not distinguishable from differences between random splits of the same twelve samples.

Within the ALPACA modules, the reference TFs were not unusually central. Their within-module weighted out-strength ranked at the 16th to 78th percentile among the module's TFs: TFAP2A 73rd (module 2), SP1 78th and NRF1 39th (module

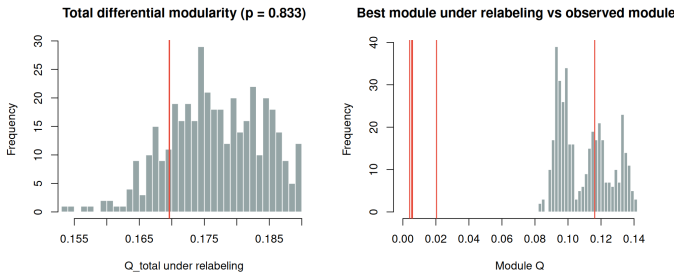


Fig. 12. ALPACA under within-timepoint FXS/control label permutation (all 399 relabelings). Left: total differential modularity Q; red line: observed ($p = 0.83$). Right: largest module Q_m in each relabeled run (grey) vs. the five largest observed modules (red); the best observed module has family-wise $p = 0.37$.

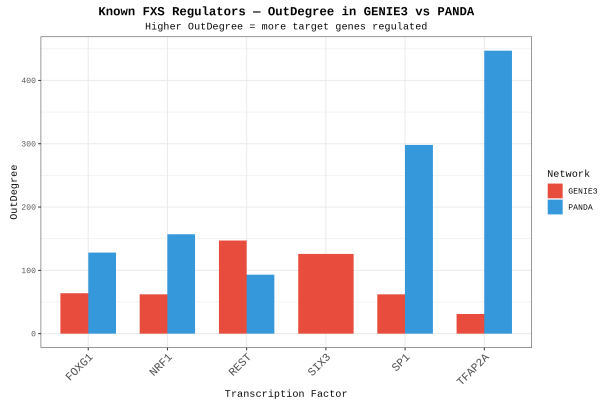


Fig. 13. Out-degree of the six reference TFs in the GENIE3 and PANDA networks. None is a genome-wide hub except REST in GENIE3 (rank 29 of 1,330).

8), REST 64th (module 7), FOXG1 39th (module 3) and SIX3 16th (module 16). Their relevance is therefore not visible in network position alone (Fig. 13).

F. Power

The power analysis shows what these tests could have detected (Table II). For LIONESS, an edge needs an FXS–control difference of 5.7 times its across-sample standard deviation (median over edges) to be detected with 80% power if 100 edges carry true effects, and 8.7 times for a single edge; no observed edge difference reached its MDE. With the same per-edge variance, the MDE would fall to 2.9, 1.8, 1.1 and 0.8 standard deviations at 12, 24, 50 and 100 samples per group. For GENIE3, none of the 300 spiked edges reached FDR < 0.05. This is not a matter of null resolution: no spiked edge came near the floor of 2.4×10^{-8}, the smallest spiked p being 2.8×10^{-6} (at $r = 0.99$). Even at $r = 0.99$ the spiked TF ranked first among 1,330 candidate regulators for only 28% of targets (top five: 62%): with 12 samples, many TFs correlate with any target by chance, and GENIE3 spreads importance across them. For ALPACA, with 20 relabelings per setting (smallest attainable $p = 0.048$, so a spike is detected only if it exceeds every relabeling), no spike was detected, from 900 edges (0.03% of the network) at δ up to 4 to 90,000 edges (3.1%) at $\delta = 2$ (one standard deviation $= 1.55$). Total

TABLE II
WHAT THE NETWORK TESTS CAN DETECT AT SIX VERSUS SIX

Test	Simulated or computed effect	Detected
LIONESS (analytic)	80% power, 100 true edges	needs 5.7 SD
	same, 50 per group	needs 1.1 SD
GENIE3 (spike-in)	$r = 0.5$–0.99, 50 edges each	0 of 300
	true TF ranked 1st at $r = 0.99$	28%
ALPACA (spike-in)	900 edges, $\delta = 0.25$–4	0 of 6
	18,000 edges, $\delta = 1, 4$	0 of 2
	90,000 edges, $\delta = 2$	0 of 1 ($p = 0.19$)

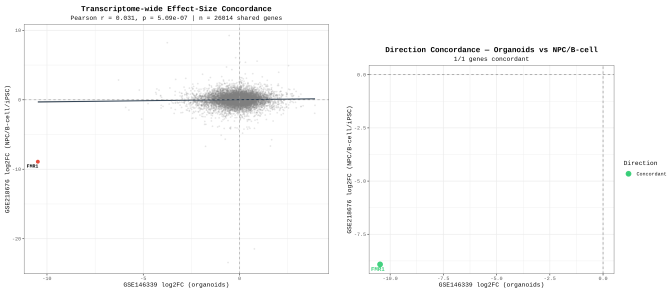


Fig. 14. Cross-cohort concordance. Left: genome-wide effect sizes across 26,014 shared genes ($r = 0.031$; sign concordance 50.5%). Right: direction of change for the shared DEG, *FMR1* (concordant, 1/1).

differential modularity did respond to the largest block (0.170 to 0.183; $p = 0.19$), confirming that the spike reached the test, but the block was not recovered as a module (22% of its nodes in the strongest module). Because ALPACA scales the FXS network to the control network's total weight and drops edges that are negative in the control network, it registers only changes that reshape a large share of the network.

G. Cross-Cohort Validation

Genome-wide effect sizes barely correlated between cohorts (Pearson $r = 0.031$, $p = 5.1 \times 10^{-7}$; Spearman $\rho = 0.020$), and sign concordance was 50.5%, indistinguishable from chance (Fig. 14). Shrinkage did not change this ($r = 0.008$, $p = 0.19$, shrunken $\log_2$FC), so the weak agreement is not an artefact of noisy low-count estimates. RRHO nevertheless detected a concordant region ($-\log_{10} p = 13.53$), located at large rank cutoffs in the body of the ranked lists rather than among top-ranked genes (Fig. 15). Only *FMR1* was shared between the primary (13) and validation DEG sets (21 of the 63 testable in both cohorts), concordantly downregulated; given the small set sizes this single-gene overlap is 95× the chance expectation (hypergeometric $p = 0.010$; Fisher OR $= 108$).

H. Tissue-General versus Tissue-Specific Effects

Only *FMR1* and *FMR1-AS1* were tissue-general under the ≥2-of-3 rule; *FMR1* was significant in all three contexts, and *FMR1-AS1* in the organoid cohort and the methylation cohort through the shared promoter probes. Of 90 genes significant in exactly one context, 11 were specific to the organoid cohort, 60 to the validation lines and 19 to blood methylation (Fig. 16). Each pairwise overlap exceeded chance (organoid–blood: 2 genes vs. 0.009 expected, $p = 3.8 \times 10^{-5}$), but

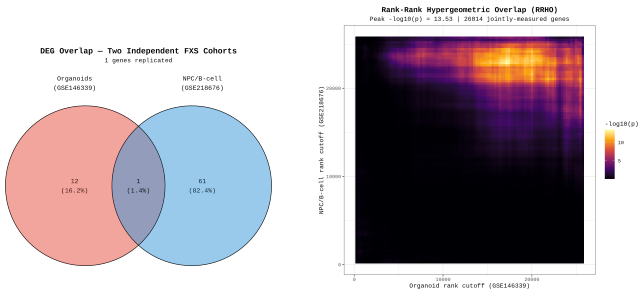


Fig. 15. Left: DEG overlap between cohorts (only *FMR1*). Right: RRHO heatmap (26,014 genes ranked by signed $-\log_{10} p$; 99×99 grid; peak $-\log_{10} p = 13.53$).

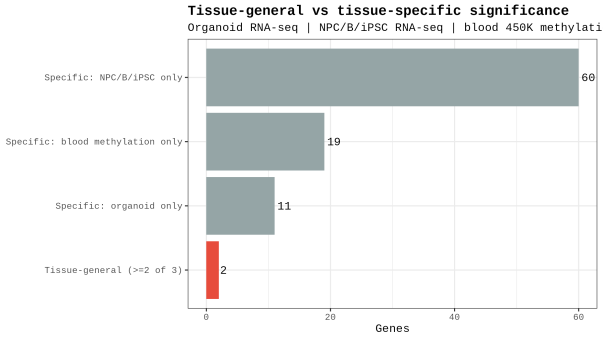


Fig. 16. Tissue-generality classification across organoid RNA-seq, NPC/B-lymphocyte/iPSC RNA-seq and blood methylation. Tissue-general: *FMR1*, *FMR1-AS1*.

was carried entirely by the *FMR1* locus. The classification was unchanged when validation DEGs came from the donor-blocked model. Together with the near-zero genome-wide correlation, this is consistent with downstream change being largely tissue-restricted, with the causal locus as the exception; with one or two small cohorts per tissue, however, it cannot be distinguished from limited power to detect shared effects.

I. In Silico Regulator Validation

ENCODE ChIP-seq self-recovery was evaluated for 22 TF–network pairs (11 TFs, GENIE3 and PANDA targets each). REST ranked first for both GENIE3- and PANDA-derived targets, but random 200-gene sets from the same universe also ranked REST highly (median rank 4; first in 27% of sets; $p = 0.29$ for GENIE3 and 0.26 for PANDA), so this rank reflects the composition of the gene universe rather than network inference. SPI1, NRF1 and BHLHE40 (PANDA; ranks 2–4) ranked better than every random set, but their targets from the motif prior alone ranked equally well (2, 2 and 3), so they reflect the prior. Only MAZ (GENIE3, rank 7) beat every random set and every shuffled network, which, with 22 pairs tested and a smallest attainable p of 0.03, is not distinguishable from chance (Fig. 17).

Curated-target enrichment uses the network's own genes as background and is therefore not affected by the gene universe. In TRRUST, one of 24 pairs passed FDR: REST with GENIE3 targets ($4.6\times$ enrichment, 8 of 22 curated targets, FDR $= 0.004$). CollecTRI, an independent and larger curated resource, gave the same result ($4.3\times$, 17 of 50 curated targets,

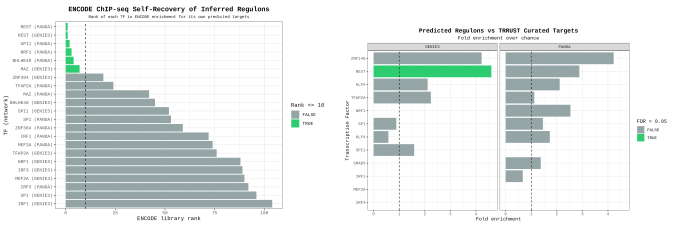


Fig. 17. In silico validation. Left: ENCODE ChIP-seq self-recovery rank (green: rank ≤ 10); random gene sets from the same universe rank REST comparably, so these ranks are not specific to the networks. Right: fold enrichment of predicted targets among TRRUST curated targets (green: FDR < 0.05).

FDR $= 5.1 \times 10^{-6}$). This enrichment depends on the real co-expression: REST's top targets in the five shuffled GENIE3 networks were not enriched (best $p = 0.20$). The weaker PANDA enrichments are circular: REST, SP1 and TFAP2A targets taken from the motif prior alone were at least as enriched in CollecTRI ($p = 0.0004$, 0.003 and 0.001) as PANDA's own ($p = 0.005$, 0.014 and 0.045).

TF activities inferred from CollecTRI regulons did not correlate between cohorts (737 TFs; Spearman $\rho = -0.019$, $p = 0.60$; Fig. 18). With the DESeq2 Wald statistics, 38 TFs were significant in the organoids and 33 in the validation lines, and one in both: CTNNB1, fewer than the 1.7 expected by chance for independent lists of these sizes ($38 \times 33/737$). Because two donors contribute two lines each to the validation cohort (donor correlation 0.66), activity was re-estimated from the donor-blocked limma-voom statistics: 11 TFs remained significant there, none of them also significant in the organoids (0.6 expected by chance). CTNNB1 was lower in FXS in both cohorts (organoid score -3.89, FDR $= 0.004$; validation -3.10, FDR $= 0.07$ donor-blocked, 0.013 unblocked), as were SP1 (organoid -2.45, FDR $= 0.14$; validation -2.82, FDR $= 0.12$ donor-blocked, 0.003 unblocked) and TFAP2A (organoid FDR $= 0.22$; validation FDR $= 0.13$ donor-blocked). SIX3 was reduced in organoids only (FDR $= 0.003$), and REST activity did not change in either cohort.

VI. Discussion

These results support a narrower and more specific claim than the framework was built to test. *FMR1* silencing replicates across three datasets and three tissues, and its promoter hypermethylation is robust to blood-composition adjustment; this is expected for the causal lesion. What does not replicate is genome-wide regulatory rewiring: cross-cohort effect-size correlation is near zero with or without shrinkage, TF activities do not correlate between cohorts, no GENIE3, PANDA or LIONESS edge passes FDR, PANDA's strongest edges are no stronger than those from a randomized prior, and ALPACA finds no module that differs between FXS and control more than between random splits of the same samples. These nulls are calibrated and can reach FDR < 0.05 (for GENIE3, once four or more edges reach the null floor), but the power analysis shows how little they can exclude: at six versus six, LIONESS detects only edge differences of about six standard deviations, GENIE3 cannot single out even a near-

TABLE III
SUPPLEMENTARY MATERIALS. ALL PATHS ARE RELATIVE TO HTTPS://GITHUB.COM/ADJ-04/FXS-RESEARCH.

Item	Content	Repository file(s)
S1	Primary cohort DESeq2 results (13 DEGs)	`results/DE/DESeq2_FXS_vs_Control_all.csv`, `DESeq2_significant_DEGs.csv`
S2	Validation cohort DE: DESeq2 (63 DEGs) and donor-blocked limma-voom	`results/validation/GSE218676_validation_DEA_all.csv`, `GSE218676_validation_limma_donorblock_all.csv`, `validation_donor_sensitivity.csv`
S3	Methylation: DMPs, gene summary, cell proportions, model comparison, *FMR1* probes	`results/methylation/DMP_FXS_vs_Control_significant.csv`, `methylation_gene_summary.csv`, `EpiDISH_cell_proportions.csv`, `DMP_model_comparison.csv`, `FMR1_probes_celltype_sensitivity.csv`
S4	Promoter methylation per gene (PANDA prior weight)	`results/methylation/promoter_methylation_by_gene.csv`
S5	GENIE3 network, centrality, target universe	`results/GENIE3/`
S6	PANDA network, centrality	`results/PANDA/`
S7	LIONESS bootstrap and top limma-tested edges	`results/LIONESS/`
S8	ALPACA modules and node scores (control reference), label-permutation runs, module family-wise p, local centrality	`results/robust_nulls/ALPACA_*`
S9	Cross-cohort concordance, overlap test, RRHO matrix	`results/validation/crossvalidation_*.csv`, `RRHO_manual_matrix.csv`
S10	ENCODE (ChEA3), TRRUST and CollecTRI validation; TF activity (DESeq2 statistics)	`results/validation/`
S11	Tissue-generality classification and pairwise tests	`results/tissue_generality/`
S12	Cytoscape network, style and session	`networks/`
S13	Scripts, Colab notebook, figures, logs, headline numbers	`scripts/`, `colab/`, `figures/`, `logs/`, `results/v5_headline_numbers.csv`
S14	GENIE3 matched-tree null with calibration; PANDA randomized-prior and shuffled-expression nulls	`results/robust_nulls/GENIE3_*`, `PANDA_*`, `part5_summary.csv`
S15	Power analysis (LIONESS MDE, GENIE3 and ALPACA spike-ins); donor-blocked TF activity; ChEA3 and CollecTRI negative controls	`results/review_checks/`

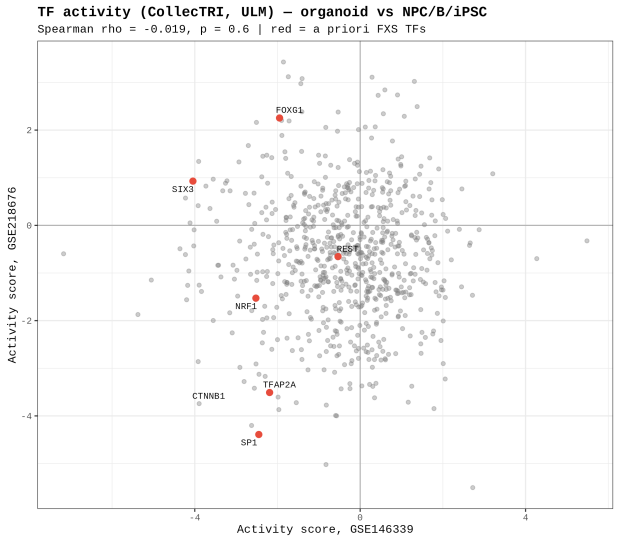


Fig. 18. TF activity (CollecTRI regulons, univariate linear model) in the organoid vs. validation cohort. Red: reference TFs. CTNNB1 is significant in both cohorts with the DESeq2 statistics shown, but not in the validation lines once donor is modelled (FDR = 0.07).

perfectly co-expressed regulator among 1,330 candidates, and ALPACA misses a two-standard-deviation shift across 3% of the network. The negative results therefore rule out only very large network changes in these samples; they are not evidence that FXS leaves regulatory networks unchanged. The tissue-generality classification is consistent with downstream effects being largely tissue-specific, with the *FMR1* locus the sole shared signal, but with one or two small cohorts per tissue we cannot distinguish tissue restriction from insufficient power to detect shared effects. The validation cohort adds a caution: once donor structure is modelled, its 63 line-level DEGs reduce to *FMR1* alone, so the true number of tissue-specific changes is uncertain.

Not everything is negative. The shuffled-expression null shows that the organoid expression data do reshape the PANDA network, moving it measurably away from its priors; the change is spread across many edges rather than concentrated in a few strong ones, which at this sample size cannot be resolved edge by edge. Among the a priori regulators, REST has the most robust support, from one kind of evidence: its GENIE3-predicted targets are enriched about four-fold among curated REST targets in both TRRUST and CollecTRI, and the enrichment disappears when GENIE3 is run on shuffled expression, so it reflects co-expression in the organoid data. Two further lines of apparent support did not survive negative controls: REST's first rank in ENCODE ChIP-seq recovery is matched by random gene sets from the same universe, and

its PANDA enrichment is matched by the motif prior alone. REST (NRSF) represses neuronal genes in non-neuronal and progenitor cells, which makes a recoverable REST regulon plausible in a neurodevelopmental model. REST activity, however, did not differ between FXS and controls, so the evidence is that the networks recover REST's regulon, not that REST is dysregulated in FXS. SP1 and CTNNB1 activities were lower in FXS in both cohorts, but once donor structure is modelled neither reaches FDR < 0.05 in the validation lines (FDR 0.12 and 0.07), and SP1 is not significant in the organoids either. The single TF significant in both cohorts with the unblocked statistics is fewer than the 1.7 expected by chance, so the cross-cohort agreement on CTNNB1, the transcriptional effector of canonical Wnt signalling, is consistent with a chance overlap; it and SP1 are exploratory observations only.

The ALPACA analysis shows why the choice of null matters. Because ALPACA normalizes node scores within each module, a module's mean score is $2/\text{size}$, so small modules look exceptional against any pooled node null by construction. Scored by raw differential modularity and tested against label permutations, which keep the dependence structure of the data intact, no module stands out. Likewise, a GENIE3 null built with fewer trees than the observed network is wider than it should be and miscalibrated; matching the tree count and checking calibration on held-out null replicates removes that ambiguity. Validation needs the same care: without controls that share the gene universe and the priors, the ENCODE ranking and the PANDA enrichments would have been read as independent support. Reporting module-, hub- or regulator-level findings without such checks, the donor-blocked reanalyses, or the cell-composition adjustment would have overstated what these data support.

Finally, the significant RRHO peak alongside chance-level sign concordance is a general lesson for small-cohort genomics: rank-based overlap and aggregate correlation measure different things, and neither alone shows whether two cohorts agree. A second lesson is methodological: per-edge permutation nulls with tens of permutations cannot support FDR control over millions of edges, and should be replaced by pooled nulls or many more permutations.

VII. Limitations

As with most network studies of rare disorders, the scope of these results is shaped mainly by the available data. The cohorts are small (six FXS and six control organoids; five lines per group in validation), so the absence of edge-level structure should be read as a statement about this scale rather than about FXS regulation in general. The three datasets come from different tissues and individuals, which enables the tissue-generality comparison but means a gene classed as tissue-specific may simply be under-detected elsewhere; for the same reason, methylation contributes a population-level prior rather than subject-matched integration, and age and array batch, which are not recorded for GSE41273, could not be modelled. Timepoint is handled as an additive effect, so FXS effects that differ between developmental stages are not captured, and the variance-ranked target universe may omit some low-variance targets. The pooled edge nulls assume scores are comparable across edges, and the shuffled-expression null measures PANDA's response to co-expression structure rather than sampling uncertainty, so it is not used to call individual edges. The ALPACA spike-ins used 20 relabelings and one to three replicates per setting, so they locate rather than precisely estimate its detection limit, and the LIONESS projections to larger cohorts assume unchanged per-edge variance. The LIONESS MDE describes the limma edge test used here, which is conservative (its p-values show a deficit near zero, and the twelve sample networks are not independent because each is derived from the same aggregate network); a label-permutation calibration of the edge test could have somewhat higher power. Finally, agreement with TRRUST and CollecTRI supports the plausibility of the inferred REST regulon; establishing causal roles in FXS will require experimental follow-up.

VIII. Future Work

The most direct next step is applying this framework to larger FXS cohorts as they become available; the power analysis suggests about 50 samples per group before edge-level differences of one standard deviation become detectable. Single-cell and spatial transcriptomics would allow cell-type-specific rewiring to be resolved, matched multi-omic cohorts would allow methylation to enter at the subject level, and interaction models would capture timepoint-specific FXS effects. Methodologically, comparison with ARACNe and GRNBoost2 and sensitivity analyses across target-universe sizes and network thresholds would show how much of the reported structure is method-specific. Experimentally, the REST regulon, reduced SP1 and CTNNB1 activity are the most specific candidates for perturbation (CRISPRi, reporter assays, ChIP) in FXS neural models.

IX. Conclusion

We built a reproducible GRN framework for Fragile X syndrome that combines RNA-seq, cell-composition-adjusted methylation, GENIE3, PANDA, LIONESS and ALPACA with calibrated permutation nulls for every network statistic, covariate-adjusted sample-specific networks, a second independent cohort, a tissue-generality classification, three-source in silico validation with negative controls, and a power analysis. *FMR1* silencing replicates across all datasets, confirming known biology. At the sample sizes available, no edge, module or genome-wide regulatory shift clears significance, and the power analysis shows that only very large effects could have. Shared downstream change across tissues is limited to the *FMR1* locus. The expression data measurably reshape the inferred networks, GENIE3 recovers the REST regulon in a way that survives shuffled-data controls, while lower SP1 and CTNNB1 activity in FXS in both cohorts remains exploratory. We release the pipeline, a Colab notebook that reproduces it from public data, all result tables including negative results, and run logs as a template for regulatory network analysis in small rare-disease cohorts.

Data and Code Availability

Raw data are public in NCBI GEO (GSE146339, GSE218676, GSE41273) and are fetched by `scripts/01_`

`fetch_data.R`. Code, result tables (including negative-result and permutation-null tables), network files, figures, Cytoscape style and session files, package versions and full run logs are at https://github.com/adj-04/fxs-research; `colab/FXS_GRN_v5_Colab.ipynb` reproduces the analysis on Google Colab. Very large intermediate tables (complete GENIE3, PANDA and LIONESS edge lists) are regenerated by the scripts. Table III maps supplementary items to repository files.

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